

PROJECT PROPOSAL

Smart Campus Application

Software Design & Analysis

SUBMITTED BY

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Submitted To:

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1. Project Title

Smart Campus Application — An Integrated University Management System

2. System Description

The Smart Campus Application is a multi-role software platform designed to digitalize the core operations of a university campus. The system provides a unified interface for Students, Teachers, and Administrative Staff (Finance, Accounts, Registrar, Examination Department, and Security).

Key features include QR-based attendance tracking, fee management and payment, examination scheduling and results, facility booking, campus announcements, a complaint and lost-and-found system, and a security and emergency alert module. The system is designed around object-oriented principles and supports role-based access for all user types.

3. Problem Statement

Universities currently operate with fragmented, department-specific tools that create significant inefficiencies. Students must navigate multiple disconnected portals for attendance, fees, exam results, and campus resources. Administrative departments — Finance, Accounts, Registrar, and Exam — rely on manual and siloed processes causing delays, errors, and miscommunication.

Specific problems include:

- No centralized platform connecting students, faculty, and all administrative offices
- Manual, error-prone attendance tracking with no real-time visibility
- Students lack timely awareness of fee dues, deadlines, and payment status
- Disconnected exam and result management with no digital rechecking workflow
- No digital system for facility reservations or campus resource management
- Clearance certificates require students to physically visit multiple offices
- No unified emergency alert or campus security communication system

The Smart Campus Application solves these problems by providing an integrated, role-based platform that automates workflows and enables real-time communication across the entire campus.

4. Objectives

1. Develop a centralized, multi-role platform accessible to all university stakeholders.
2. Implement QR-code and GPS-based digital attendance tracking.
3. Provide a real-time fee management module with automated challan generation and payment alerts.
4. Build a digital examination module for scheduling, results, and rechecking workflows.
5. Design a facility booking system with admin approval and QR-based access.
6. Enable targeted campus-wide announcements and push notifications.
7. Create a complaint and lost-and-found system with smart item-matching.
8. Implement a multi-department clearance certificate workflow.
9. Develop a security module with gate logs, visitor management, and emergency broadcasts.
10. Demonstrate OOP principles (inheritance, encapsulation, polymorphism, association) throughout the system design.
11. Produce complete and consistent UML documentation covering all diagram types studied in the course.

5. Software Design Specification (SDS) Outline

5.1 Use Case Diagram

Captures all system actors and their interactions with the system's functional use cases.

Actor	Key Use Cases
Student	Login, View Timetable, Scan QR Attendance, Pay Fee, View Results, Book Facility, Submit Complaint, Request Clearance
Teacher	Login, Generate QR, Mark Attendance, Submit Grades, Post Announcement
Finance Dept.	Generate Fee Challan, Track Payments, Manage Scholarships, Send Reminders
Accounts Office	Verify Payments, Approve Clearance, View Audit Logs
Exam Dept.	Schedule Exams, Publish Results, Process Rechecking, Issue Transcripts
Registrar	Manage Enrollment, Approve Course Registration, Issue Certificates
Security Staff	Log Gate Entry/Exit, Register Visitor, Trigger Emergency Alert
Admin	Manage All Users, Approve Facility Bookings, View Reports

Notable relationships:

- Pay Fee <<includes>> Verify Payment
- Request Clearance <<includes>> Check All Department Approvals
- Scan QR Attendance <<extends>> Verify GPS Location
- Submit Complaint <<extends>> Escalate to Admin (if unresolved after 72 hrs)

5.2 Class Diagram

Models the system's entities, attributes, methods, and relationships.

Class	Key Attributes	Key Methods
User (abstract)	userID, name, email, role	login(), logout(), sendNotification()
Student	rollNo, program, semester	viewAttendance(), viewFee(), requestClearance()
Teacher	employeeID, department	generateQR(), submitGrades(), postAnnouncement()
Course	courseCode, title, creditHours	getEnrolledStudents(), getAttendance()
AttendanceRecord	studentID, courseID, date, status	markPresent(), markAbsent(), getPercentage()
FeeRecord	studentID, totalAmount, status, dueDate	generateChallan(), updateStatus()
Payment	paymentID, amount, method, date	verify(), generateReceipt()
ExamSchedule	courseID, date, venue, invigilator	generateSeatingPlan()
Result	studentID, courseID, marks, grade	publish(), applyRechecking()
FacilityBooking	studentID, facilityID, slot, status	submit(), approve(), generateQR()
Clearance	studentID, status, approvals[]	checkAllDepts(), generateCertificate()
Notification	recipientID, type, message	send(), markRead()
EmergencyAlert	type, triggeredBy, affectedZone	broadcast()

OOP relationships in the class diagram:

- Inheritance — Student, Teacher, StaffMember all extend User
- Association — Student enrolled in Course; Teacher teaches Course
- Aggregation — Department contains many Courses
- Composition — FeeRecord composed of FeeLineItems; Clearance composed of DepartmentApprovals

5.3 Activity Diagrams

Illustrate the workflow and decision logic within key system processes.

- QR Attendance Flow — QR generation, student scan, GPS check, attendance recording, low-attendance alert
- Fee Payment Flow — challan generation, student payment, accounts verification, receipt issuance, overdue flagging
- Clearance Certificate Flow — student request, parallel checks across Finance/Library/Registrar, certificate issuance
- Facility Booking Flow — student request, admin approval/rejection, confirmation, QR access at booking time
- Exam Result Flow — teacher submits marks, exam dept review, result publication, rechecking request handling

5.4 Sequence Diagrams

Show time-ordered message exchanges between objects for key system interactions.

- QR Attendance — Teacher, QRGenerator, StudentApp, GPSVerifier, AttendanceDB, NotificationService
- Fee Payment — Student, FeeController, PaymentGateway, AccountsOffice, ReceiptService
- Clearance Request — Student, ClearanceController, FinanceDept, LibrarySystem, RegistrarOffice, AccountsOffice
- Emergency Alert — SecurityStaff, AlertController, NotificationBroadcaster, AllUsers, CampusMapService

5.5 Collaboration Diagrams

Show structural object relationships with numbered message flows between collaborating objects.

- Attendance System Collaboration — Teacher, QRCodeGenerator, AttendanceRecord, Student, NotificationService
- Payment Verification Collaboration — Student, FeeRecord, PaymentGateway, AccountsOffice, Receipt
- Emergency Alert Collaboration — SecurityStaff, AlertSystem, NotificationBroadcaster, UserRegistry, CampusMap